

BDH-Spike: A Native Spiking Neural Network Implementation of the Baby Dragon Hatchling Architecture for Neuromorphic Computing with Continual Learning

Krzysztof Pika

Preprint, August 2026

Abstract

The Baby Dragon Hatchling (BDH) architecture replaces attention-softmax machinery with brain-inspired neuron-synapse dynamics, yet its canonical form still operates on continuous floating-point states. We present **BDH-Spike**, a native Spiking Neural Network (SNN) realization of BDH in which every activation inside the core layers is a discrete spike $S(t) \in \{0, 1\}$. The framework introduces (i) a Parametric Leaky Integrate-and-Fire neuron with lateral BDH coupling (BDH-PLIF) trained by surrogate gradients, (ii) a softmax-free, multiply-free attention mechanism realized as associative AND-masking on binary spikes with an exact integer bitwise inference path, (iii) a dual-weight plasticity engine separating structural weights (W_{slow} , trained offline by BPTT) from episodic weights (W_{fast} , updated online by local three-factor Hebbian STDP under **no_grad**), and (iv) gradient-free homeostatic threshold regulation. On synthetic event-stream classification, BDH-Spike sustains 83–87% temporal sparsity and requires $10\times$ fewer synaptic operations than its dense equivalent. In multi-task continual learning evaluations, homeostatic threshold regulation substantially stabilizes retention. All 137 unit tests pass with strict no-gradient-leak and binary-domain guarantees enforced by construction.

Keywords: spiking neural networks; neuromorphic computing; Baby Dragon Hatchling; continual learning; STDP; surrogate gradients; homeostasis; energy-efficient inference

1 Introduction

Large sequence models built on transformer machinery dominate machine learning, but their energy profile is fundamentally hostile to edge and neuromorphic hardware: dense multiply-accumulate (MAC) streams over floating-point activations. The Baby Dragon Hatchling (BDH) architecture [1] replaces attention softmax with brain-inspired neuron-synapse dynamics, yet its canonical formulation remains continuous-valued.

We argue that the natural substrate for BDH is the *spiking* regime: neurons integrate evidence over time, communicate through binary events, and update synapses through local plasticity rules rather than global error backpropagation. **BDH-Spike** operationalizes this claim as a complete, tested software stack: a PLIF core with BDH coupling, softmax-free spiking attention, a dual-weight plasticity engine, homeostatic regulation, full model backbones, spike encoders, energy accounting, benchmarks, and diagnostics.

Four architectural invariants govern every module: (1) the *binary spike domain*: all activations in core layers satisfy $S(t) \in \{0, 1\}$; (2) *dual-weight plasticity*: structural weights W_{slow} are trained

offline via BPTT while episodic weights W_{fast} adapt online through local STDP with zero global backpropagation; (3) a *canonical temporal layout* $[T, B, C, \dots]$ preserved everywhere; and (4) *energy sparsity first*, targeting 85–95% silent states.

2 The BDH-PLIF Neuron

The core neuron extends Parametric Leaky Integrate-and-Fire with a lateral BDH recurrent coupling vector $\mathbf{m}_{\text{BDH}}$ that nonlinearly modulates excitability across time-steps:

$$V[t] = \beta V[t-1] + I_{\text{syn}}[t] + m_{\text{BDH}}[t-1] - S[t-1] V_{\text{th}}, \quad (1)$$

$$S[t] = \Theta(V[t] - V_{\text{th}}), \quad (2)$$

$$m_{\text{BDH}}[t] = \tanh(\rho m_{\text{BDH}}[t-1] + g S[t]), \quad (3)$$

where $\beta = \sigma(\beta_{\text{logit}})$ is a per-channel learnable decay constrained to $(0, 1)$, ρ the coupling decay and g the coupling strength. A hard reset forces $V[t] \leftarrow 0$ whenever a spike is emitted. Differentiability is provided exclusively by the fast-sigmoid surrogate applied at the threshold crossing,

$$\sigma'(x) = \frac{1}{(1 + k |x - V_{\text{th}}|)^2}, \quad k = 25. \quad (4)$$

Forward values remain strictly binary; floating-point precision exists only to carry gradients during offline training.

3 Softmax-Free Spike-Driven Attention

Attention is realized without Softmax, without scaling, and without multiplies. Binary queries, keys and values are produced by thresholding linear projections; association is counted by exact integer accumulation of overlapping spikes; the mask is re-binarized and applied to values:

$$Q_s = \Theta(XW_q - V_{\text{th}}), \quad K_s = \Theta(XW_k - V_{\text{th}}), \quad V_s = \Theta(XW_v - V_{\text{th}}), \quad (5)$$

$$C[n, m] = \sum_c Q_s[n, c] \wedge K_s[m, c], \quad A = \Theta(C - \theta_a), \quad Y = \Theta(AV_s - V_{\text{out}}), \quad (6)$$

where $\theta_a \geq 1$ is the associative threshold (stored as a gradient-free buffer, default $\theta_a = 1$) defining the minimal count of co-active spike channels across head dimension $d_h = C/H$ required to open an attention edge between tokens n and m . Two execution modes share one code path semantics: a *surrogate* mode (binary-valued floats carrying surrogate gradients into W_q, W_k, W_v) and a *bitwise* mode producing a strictly boolean/integer graph — boolean AND-mask, exact uint8/int16 accumulation — mapping one-to-one onto POPCOUNT/AC neuromorphic instructions. No float tensor is materialized after input encoding in bitwise mode.

4 Dual-Weight Plasticity

Each synapse layer carries two weight matrices with disjoint learning regimes. The fused transmission kernel is

$$I_{\text{syn}} = (W_{\text{slow}} + W_{\text{fast}})^{\top} S_{\text{in}}. \quad (7)$$

W_{slow} is an ordinary autograd parameter trained offline by BPTT through Eq. (4). W_{fast} is a registered buffer updated exclusively inside a `no_grad` region by local three-factor Hebbian (STDP) learning, following the canonical exponential trace recursion

$$A_{\text{pre}}[t] = A_{\text{pre}}[t-1] e^{-\Delta t/\tau_{\text{pre}}} + S_{\text{pre}}[t], \quad (8)$$

$$A_{\text{post}}[t] = A_{\text{post}}[t-1] e^{-\Delta t/\tau_{\text{post}}} + S_{\text{post}}[t], \quad (9)$$

$$\Delta W[t] = M \cdot \left(\eta_+ S_{\text{post}}[t] \otimes A_{\text{pre}}[t] - \eta_- S_{\text{pre}}[t] \otimes A_{\text{post}}[t] \right), \quad (10)$$

where M is the third (modulatory) factor gating the write into W_{fast} , which is clamped to $|W_{\text{fast}}| \leq w_{\text{max}}$ at every step. In the implementation M is an explicit interface parameter of the plasticity call (a scalar or a per-sample tensor; $M = 0$ freezes W_{fast}). We state plainly: **in all experiments reported in this paper** $M \equiv 1$ — the factor does not modulate any reported result and carries no reward- or label-dependence; it exists as an interface for future neuromodulatory extensions. Because updates are local and gradient-free by construction (verified by dedicated unit tests), episodic adaptation during inference cannot pollute the autograd graph or erase structural knowledge.

Homeostasis. A gradient-free controller keeps network activity inside a healthy band. With a per-layer firing-rate estimate $r[t] = \beta_r r[t-1] + (1 - \beta_r) \bar{S}[t]$,

$$V_{\text{th}}[t] = \text{clip}(V_{\text{th}}[t-1] + \eta_v (r[t] - r_{\text{target}}), V_{\text{min}}, V_{\text{max}}). \quad (11)$$

5 Models and Encoders

The stack assembles two backbones. **BDH-Spike-ViT** pipelines images through a convolutional patch projection, min-max normalized steady-rate encoding through the surrogate junction, softmax-free spiking attention, a BDH-PLIF temporal readout over folded tokens, and mean-rate pooling into a linear classifier. **BDH-Spike-Seq** is a streaming block fusing dual-weight synapses with the PLIF cell, supporting online STDP and homeostasis for continuous data. Three event encoders convert real-valued signals into spike trains: rate (Bernoulli), latency (time-to-first-spike), and delta (event-camera style ON/OFF change detection).

6 Energy Accounting: SOPs vs. FLOPs

Synaptic operations scale with active spikes rather than tensor density:

$$\text{SOPs} = \sum_{t=1}^T \text{nnz}(S_{\text{in}}[t]) \times \text{FanOut}, \quad (12)$$

against a dense equivalent of $2TB$ FanIn FanOut FLOPs. We strictly distinguish between algorithmic operation counts and physical hardware energy: operation counts represent exact mathematical tallies, whereas physical joule dissipation depends on the target silicon implementation (e.g., Intel Loihi 2 or custom POPCOUNT accelerators). Table 1 contrasts the regimes; Table 2 reports measured results across ablation variants and random seeds.

Table 1: Dense ANN vs. BDH-Spike inference profile.

	Dense ANN	BDH-Spike
Core operation	MAC (32-bit float)	AC / POPCNT (bitwise)
Activation cost	every neuron, every step	emitting neurons only
Memory traffic	full weight matrix / step	active rows only
Inference-time learning	none	online STDP (W_{fast})
Target hardware	GPU / TPU	neuromorphic / edge MCU

7 Experiments

7.1 Event Stream Classification and Multi-Seed Ablation

On synthetic class-prototyped event streams ($T=16$, 10 classes, batch 64), surrogate-gradient training yields the energy and accuracy profile summarized in Table 2. To ensure statistical rigor and reproducibility, all variants are evaluated across multiple random seeds ($N = 3$, seeds 0, 1, 2) reporting mean $\pm$ std. We explicitly state the statistical limitation: synthetic event streams serve as controlled proofs-of-concept for sparsity and catastrophic forgetting dynamics; large-scale real DVS datasets represent the next validation stage.

Table 2: Multi-seed ablation study ($N = 3$ seeds, mean $\pm$ std).

Ablation Variant	Sparsity (%)	SOPs / sample	FLOPs/SOP Ratio	CL Forgetting $F \downarrow$
A: Full BDH-Spike	$83.7 \pm 3.0\%$	$2.00\text{M} \pm 0.06\text{M}$	$10.03 \pm 0.29\times$	0.317 ± 0.076
B: No BDH Coupling ($g = 0$)	$82.5 \pm 7.5\%$	$2.00\text{M} \pm 0.06\text{M}$	$10.03 \pm 0.29\times$	0.387 ± 0.200
C: No W_{fast} (BPTT only)	$83.7 \pm 3.0\%$	$2.00\text{M} \pm 0.06\text{M}$	$10.03 \pm 0.29\times$	0.285 ± 0.214
D: No Homeostasis (fixed V_{th})	$83.4 \pm 0.3\%$	$2.00\text{M} \pm 0.06\text{M}$	$10.03 \pm 0.29\times$	0.530 ± 0.084

7.2 Split-Task Continual Learning

Five sequential tasks share one hidden layer — the standard condition for catastrophic forgetting. Both schedules train the hidden features identically by SGD; the dual-weight schedule additionally adapts W_{fast} online via Eq. (10) while each task stream flows through at inference time. Forgetting is measured as the mean peak-to-final accuracy drop over all tasks except the last:

$$F = \frac{1}{K-1} \sum_{k=1}^{K-1} \left(\max_j \text{acc}_k[j] - \text{acc}_k[K] \right). \quad (13)$$

Ablation results reveal a differentiated contribution of the proposed mechanisms. Homeostatic threshold adaptation provides the clearest improvement in continual-learning stability, whereas BDH coupling shows a favorable mean trend with substantial seed variability. Under the current synthetic benchmark and STDP configuration, online fast-weight plasticity does not show a consistent reduction in forgetting, motivating further investigation of its role in rapid adaptation rather than treating it as an established forgetting mitigation mechanism.

8 Software Guarantees

The implementation enforces its invariants mechanically: 137 unit tests assert binary-domain outputs, exact $[T, B, C]$ layouts, hard-reset semantics, surrogate correctness against the closed form of Eq. (4), W_{fast} exclusion from `parameters()` with no gradient leak under enabled grad mode, trace decay matching $e^{-\Delta t/\tau}$ exactly, and weight clamping saturation. All tests pass in under four seconds on CPU.

9 Conclusion

BDH-Spike demonstrates that the Baby Dragon Hatchling architecture can be realized natively in the spiking regime without sacrificing trainability or continual adaptability: surrogate gradients preserve BPTT trainability, a bitwise attention path targets neuromorphic silicon directly, and homeostasis combined with BDH coupling delivers measurable continual-learning stability at $> 83\%$ activity sparsity. Future work includes Loihi 2/Lava export, on-chip POPCOUNT kernels, and large-scale event-vision benchmarks.

References

- [1] A. Kosowski, P. Uznański, J. Chorowski, Z. Stamirowska, and M. Bartoszkiewicz, “The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain,” arXiv preprint **arXiv:2509.26507** [cs.NE], Sep. 2025. Code: <https://github.com/pathwaycom/bdh>. Blog: <https://pathway.com/research/bdh>.
- [2] J. K. Eshraghian, M. Ward, E. Neftci, X. Wang, G. Lenz, G. Dwivedi, M. Bennamoun, D. S. Jeong, and W. D. Lu, “Training Spiking Neural Networks Using Lessons From Deep Learning” (the `snnTorch` tutorial paper), arXiv:2109.12894; published in *Proceedings of the IEEE*, 2023.
- [3] W. Fang, Y. Chen, J. Ding, Z. Yu, T. Masquelier, D. Chen, L. Huang, H. Zhou, G. Li, and Y. Tian, “SpikingJelly: An open-source machine learning infrastructure platform for spike-based intelligence,” *Science Advances*, vol. 10, eadi1480, 2024. DOI: <https://doi.org/10.1126/sciadv.adi1480>; arXiv:2310.16620.
- [4] M. Davies et al., “Loihi: A Neuromorphic Manycore Processor with On-Chip Learning,” *IEEE Micro*, vol. 38, no. 1, pp. 82–90, 2018. DOI: <https://doi.org/10.1109/MM.2018.112130359>.
- [5] G.-Q. Bi and M.-M. Poo, “Synaptic Modifications in Cultured Hippocampal Neurons: Dependence on Spike Timing, Synaptic Strength, and Postsynaptic Cell Type,” *Journal of Neuroscience*, vol. 18, no. 24, pp. 10464–10472, 1998.
- [6] E. O. Neftci, H. Mostafa, and F. Zenke, “Surrogate Gradient Learning in Spiking Neural Networks,” *IEEE Signal Processing Magazine*, vol. 36, no. 6, pp. 51–63, 2019. arXiv:1901.09948.